

Waste Type Identification

Machine Learning Project Work

AUTHORS

D'Errico Fabrizio · De Girolamo Del Mauro Aniello · De Vivo Manuel

ACADEMIC YEAR

A.Y. 2025/2026

Final model: ResNet18 Final — 0.9740 balanced accuracy

Architecture

- ResNet18 pretrained on ImageNet
- Head: Dropout(0.25) + Linear(512, 8)
- Two-phase fine-tuning: freeze backbone 5 epochs, then unfreeze

Training & Loss

- Loss: CrossEntropy with class weights + label smoothing 0.10
- WeightedRandomSampler for class imbalance
- AdamW, discriminative LRs (1e-4 backbone, 5e-4 head)

Optimization

- ReduceLROnPlateau + early stopping (patience 8)
- Reject option: Vento ψ_a / ψ^b , coverage $\geq 98\%$, cal/eval split
- Moderate augmentation — strong aug tested and discarded

Dataset Split & Validation Strategy

03 / 11

METHODOLOGY

80/20 StratifiedShuffleSplit

- Train: 80% (~12,400 samples) with WeightedRandomSampler
- Validation: 20% → split 50/50 into val_cal / val_eval

Why val_cal / val_eval?

- Vento et al. methodology: threshold calibrated on val_cal, evaluated on val_eval
- val_eval never seen during training or calibration — prevents data leakage

DECISION JUSTIFICATION

Why 80/20 over K-fold?

- **More training data for minority classes**
(Metal: 123 vs ~100 samples in K=3)
- **Stability confirmed by cross-validation**
(std 0.0019 across 3 folds)

🔗 Validation / Test Pipeline

Resize 256 → CenterCrop 224×224 → ToTensor →
ImageNet normalization

⚡ Training Augmentation (moderate)

- RandomResizedCrop (scale 0.80–1.0)
- RandomHorizontalFlip (p=0.5)
- RandomRotation ($\pm 10^\circ$)
- ColorJitter (brightness 0.20, contrast 0.20, saturation 0.15)
- RandomErasing (p=0.15)

CRITICAL FINDING

Why moderate?

Strong augmentation collapsed Metal precision from 0.94 to 0.82.

Aggressive transforms destroy fine-grained features of visually similar materials.

Architecture & Hyperparameters

Component	Setting
Backbone	ResNet18 (ImageNet pretrained, 11M params)
Head	Dropout(0.25) + Linear(512, 8)
Loss	CrossEntropy + class weights + label smoothing 0.10
Optimizer	AdamW, backbone LR 1e-4, head LR 5e-4, weight decay 1e-4
Scheduler	ReduceLROnPlateau (factor 0.5, patience 5)
Early stopping	Patience 8 (on balanced accuracy)
Batch size	32
Epochs	40 (5 freeze + 35 finetune)
GPU	FP32, < 5 GB training, < 4 GB inference

Experimental Plan: Key Hypotheses

06 / 11

Each experiment tests ONE hypothesis. Not a chronological sequence.

HYPOTHESIS 1

H1: Transfer learning baseline

ResNet18 full fine-tuning → 0.9487 BA. Weak on Metal (0.88), Plastic (0.89).

HYPOTHESIS 2

H2: Discriminative LRs + dropout (Regularized)

0.9487 → 0.9591. Metal recall: 0.88 → 0.92.

HYPOTHESIS 3

H3: Batch-level imbalance (Batch-Balanced)

WeightedRandomSampler + label smoothing → 0.9725 BA. Key breakthrough.

HYPOTHESIS 4

H4: Two-phase training (Final line)

Freeze backbone 5 epochs, then unfreeze. Protects pretrained weights. 0.9740 BA.

HYPOTHESIS 5

H5: Reject option (Vento et al.)

Coverage $\geq 98\%$, cal/eval split. 0.9835 accepted BA at 1.7% reject rate.

What Did NOT Work — and Why

Attempt	Result	Why Rejected
Strong augmentation	Metal P: 0.94 → 0.82	Destroys fine features of similar materials
Label smoothing 0.15	Early stop epoch 12, Metal P 0.83	Over-regularizes, weakens gradient
3-epoch freeze	BA dropped ~5 points	Head not stabilized before unfreeze
ResNet34	BA 0.9626 vs 0.9740	Deeper ≠ better for this task
SGD + momentum	BA 0.9576	Slower convergence than AdamW

Cross-Validation: Robustness Check

08 / 11

0.0019

STD ACROSS SPLITS

Mean $\pm$ std = 0.9495 $\pm$ 0.0019

Fold results: 0.9478 | 0.9486 | 0.9521

Cross-Validation: Stratified 3-fold cross-validation

- **Purpose:** measure cross-fold variance, NOT to compete on absolute performance
- **Reduced budget:** 15 epochs, 3 freeze, patience 4
- Each fold trains on 66% of data (vs 80% in final model)
- **Std 0.0019** $\rightarrow$ BA varies by $< 0.4\%$ across splits
- **Lower absolute BA expected:** less data per fold, fewer epochs
- At full budget: BA would rise, std would remain low
- **Conclusion: the 80/20 result is stable, not a lucky split**

Reject Option (Vento et al.)

09 / 11

How it works:

ψ_a = max softmax probability (confidence)
 ψ_b = top-2 probability margin (ambiguity between two classes)

- Scan ALL τ values on val_cal
- Pick τ that maximizes accepted BA with coverage $\geq 98\%$
- Evaluate on val_eval (never seen during training or calibration)
- Fully data-driven: no cost function, no arbitrary parameters

Results on val_eval:

- **No reject:**
BA 0.9740 | Coverage 100% | 36 errors
- ψ_a **(softmax):**
BA 0.9835 | Coverage 98.3% | 1.7% rejected
- ψ_b **(margin):**
BA 0.9800 | Coverage 98.9% | 1.1% rejected
- ψ_a **removes 12/36 errors at only 1.7% reject rate**

Final Results: Final

Class-wise Performance (val_eval, 1,552 samples)

Battery P=0.98 R=1.00	Metal P=0.94 R=1.00
Clothing P=0.99 R=0.99	Organic P=0.96 R=0.98
Glass P=0.97 R=0.97	Papery P=0.96 R=0.96
Plastic P=0.96 R=0.90	Undiff. P=0.97 R=1.00

Standard BA: 0.9740 | Accuracy: 0.9781 | Gap: 2.39%

With ψ_a reject: BA 0.9835 at 1.68% rejection rate

Key observations:

- **Best epoch: 40/40** — no early stop triggered. Stable convergence.
- **Metal precision (0.94)**: best across all experiments
- **Plastic recall (0.90)**: hard limit — transparent plastic vs glass = intrinsic ambiguity
- **Train/val gap (2.39%)**: lowest across all experiments — no overfitting

Take-Home Messages

11 / 11

1

Imbalance handling is critical

WeightedRandomSampler was the single most impactful change (+2.4% BA)

2

Two-phase transfer learning protects pretrained weights

Freezing the backbone prevents noisy gradients from the untrained head

3

Stronger is NOT always better

Strong aug, higher LS, deeper architectures all underperformed on this dataset

4

Reject option is a safety net with zero additional training cost

At 1.7% reject rate, accepted BA jumps from 0.9740 to 0.9835

5

Cross-validation confirms stability, not luck

Std 0.0019 across 3 folds — the 80/20 result is robust and split-independent

6

Plastic is the hard limit

Recall 0.90 across all experiments — not a tuning issue but an intrinsic dataset property